Vipul Sapra

Kharghar, Navi Mumbai | 8888-033-168 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Distributed Systems, Large Language Models, Agentic AI, Autonomous Workflows, Anomaly Detection, Predictive Modeling, Time-Series Forecasting, Unit Economics, Data Governance & Lineage, ETL/ELT Pipeline Architecture, Dimensional Modeling, MLOps, CI/CD, Cloud Lakehouses

Programming Languages & Tools: AWS (S3, Redshift, Step Functions, Glue, Kinesis), Azure (ADF, Databricks, Synapse, Azure SQL), Snowflake, Apache Spark, PySpark, Python, Scikit-Learn, XGBoost, SARIMA, Power BI, Tableau, Collibra, KIRO IDE, GitHub Copilot

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Software Dev Engineer II - AI & Profit Intelligence Platforms — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic workflows** (KIRO IDE, prompt orchestration, Copilot) to automate schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50% while supporting petabyte-scale data pipelines.
- **Engineered and deployed an autonomous AI anomaly detection framework** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, safeguarding downstream BI, pricing analytics, and data pipeline integrity.
- **Built and deployed XGBoost, Random Forests, and Scikit-Learn classification models** predicting customer behavior and unit economics at scale, delivering actionable intelligence signals that improved operational reliability and service efficiency by 20%.
- **Architected distributed cloud data platforms and dimensional models** (Star/Snowflake schemas, SCD Type 1/2) using AWS (S3, Redshift), Azure, Snowflake, and Apache Spark/PySpark to process hundreds of terabytes of incoming event data with 99.9% data availability SLAs.
- **Engineered automated data quality and governance firewalls** and Collibra data lineage tracking to ensure correct data semantics, audit usage, detect duplication, and reduce operational turnaround by over 80%.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Architected advanced time-series SARIMA predictive models** to forecast truck rolls and field support demand, optimizing workforce capacity allocation and reducing response times by 20%.
- **Engineered automated testing and evaluation framework for SpectrumIQ** enterprise conversational AI platform, cutting manual QA turnaround by 50% and accelerating production release cycles for natural language capabilities.
- **Developed scalable data ingestion pipelines and predictive analytics models** integrating disparate financial and operational telemetry sources to drive data-driven corporate strategy and automated reporting.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed and delivered executive Key Operating Metric (KOM) dashboards** and advanced semantic models in Power BI (DAX) and Tableau, translating complex transactional telemetry into actionable executive strategy.
- **Optimized enterprise reporting frameworks and data aggregation stores** to provide self-service analytics and high-throughput data access for business leaders and operational teams.
- **Implemented robust data validation and reconciliation processes** ensuring high fidelity across multi-layered data warehouse environments and automated BI reporting tools.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.